

MASTER'S THESIS DEFENCE

CVA6 Optimization

Microarchitectural optimization of the open-source RISC-V CVA6 core for
higher frequency and lower memory latency

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Introduction & Motivation

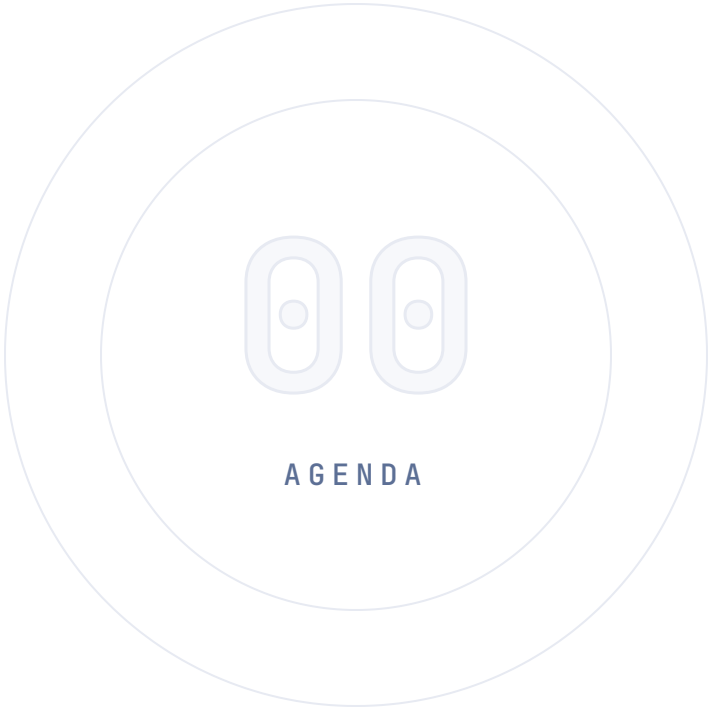
- Modern System-on-Chips (SoCs) require constant optimization across multiple layers to achieve high efficiency.
- Hardware vendors continuously innovate to keep up with the demand pace with proprietary cross platform optimization techniques and commercial protection.
- the open-source platforms seems comparatively limited. To have the comparative open-source solution, this research leverages the open-source CVA6 CPU core.



AGENDA

Outline

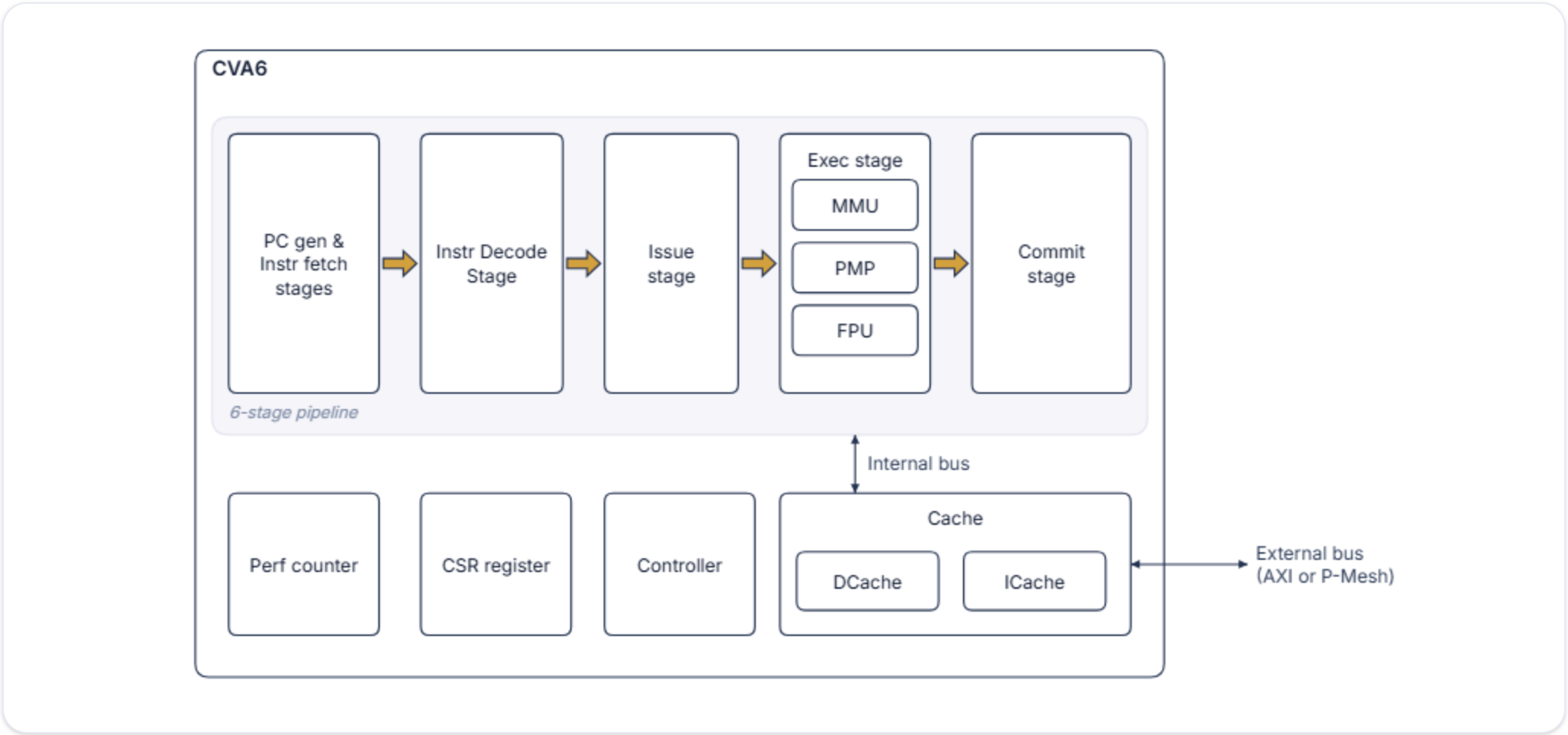
- Background: The CVA6 Core
- Analysis
- Proposed Optimizations
- Methodology & Implementation
- Results & Evaluation
- Conclusion & Future Work



— BACKGROUND

Background: The CVA6 Core

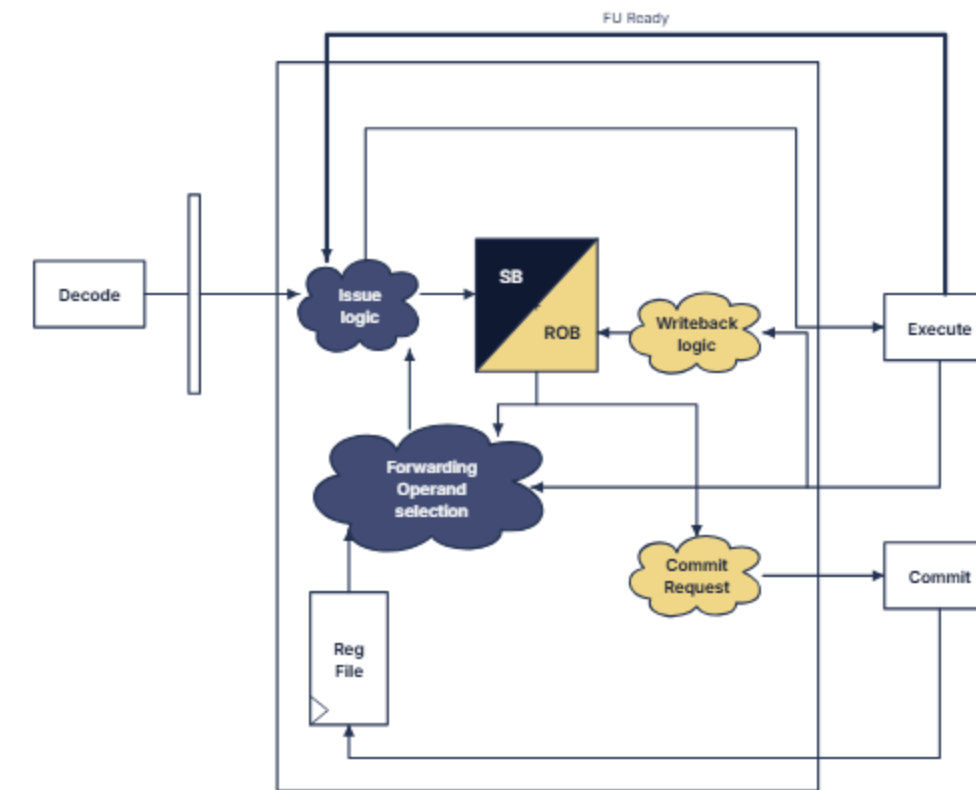
- Highly Configurable 32/64 bit RISC-V application core: CV32A6 and CV64A6
- Initiated from ETH Zurich, now curated by OpenHW
- Supports various RISC-V modular extension (I, M, F, A, C)
- 6-Pipeline stage:
 - PC generation
 - Decode (various extension decoders)
 - Ex stage (ALU, LSU, FPU, ...)
 - Frontend
 - Issue stage (SB and IRO)
 - Commit
- HPD cache in the CVA6, is a non-blocking cache (hit under miss)



ANALYSIS

Analysis

- The Scoreboard (SB) and the Controller are the largest combinational modules on the critical timing path.
- A centralized scoreboard thus high routing complexity.
- Additionally, huge penalty for cache miss penalties (the "Memory Wall") lower core utilization during data-intensive tasks.



PROPOSED OPTIMIZATIONS

Proposed Optimization Overview

- **Goal 1:** Significantly improve the processor's capability to operate at higher frequencies.
- **Goal 2:** Enhance per-cycle performance by hiding the memory latency.

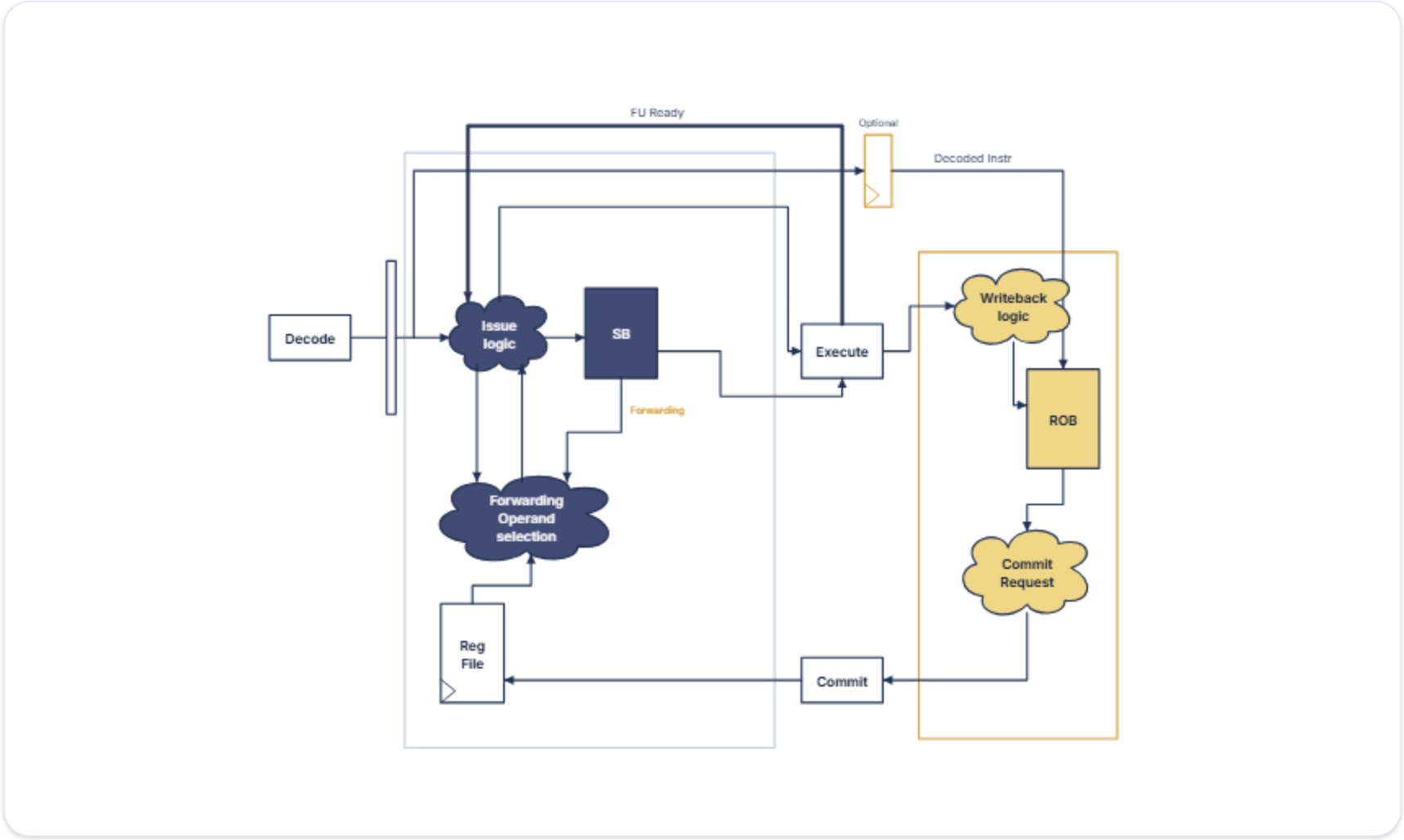


PROPOSED OPTIMIZATIONS

Microarchitecture Optimization: Scoreboard & ROB

- Introduction of new mem struct
- Distribution of responsibilities

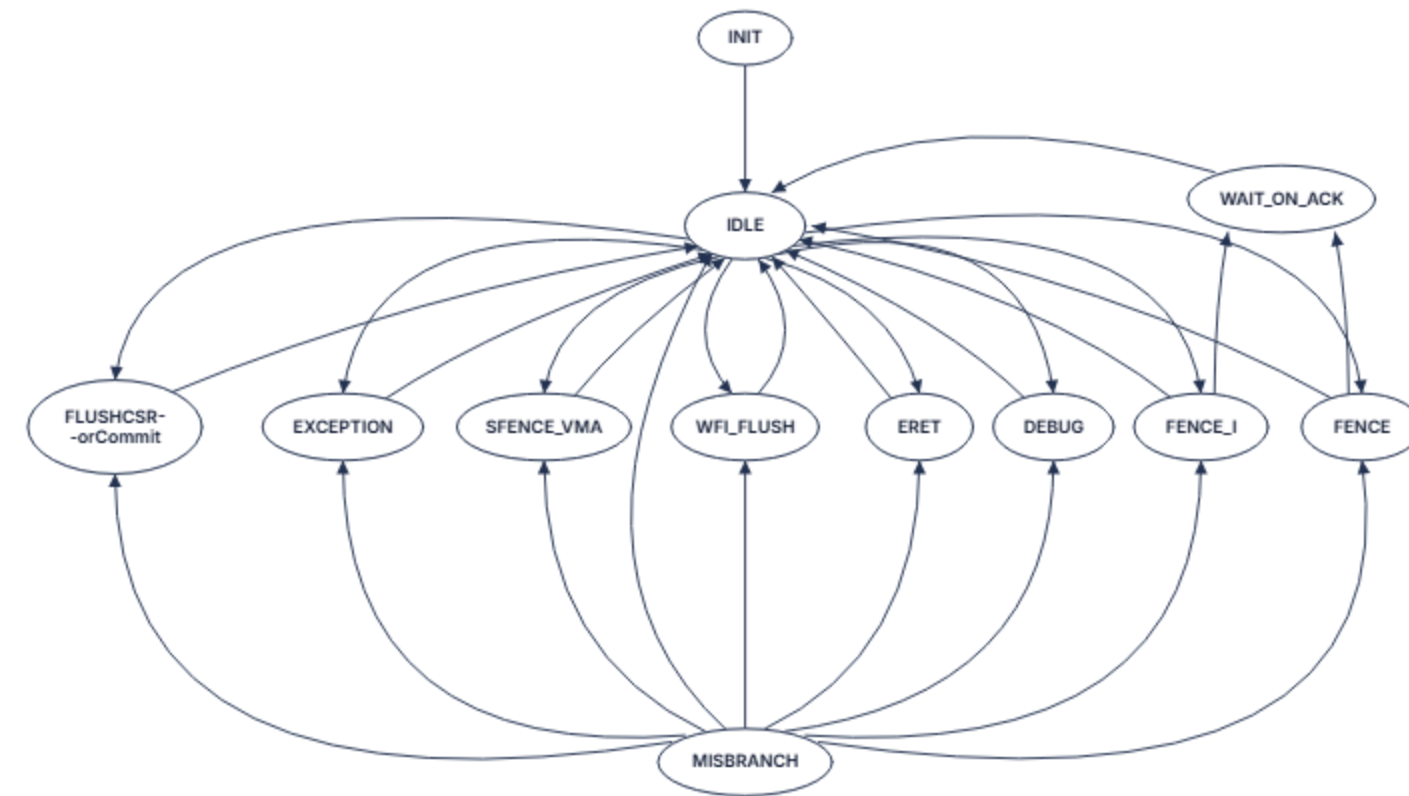
FIELD	MEANING
issued	instruction is issued
canceled	instruction is canceled due to speculative scoreboard
forwardable	result can be forwardable
rd	register destination address
result	result of the instruction
result_valid	validity of result
is_rd_fpr_flag	if rd is floating point register



PROPOSED OPTIMIZATIONS

Microarchitecture Optimization: Controller FSM

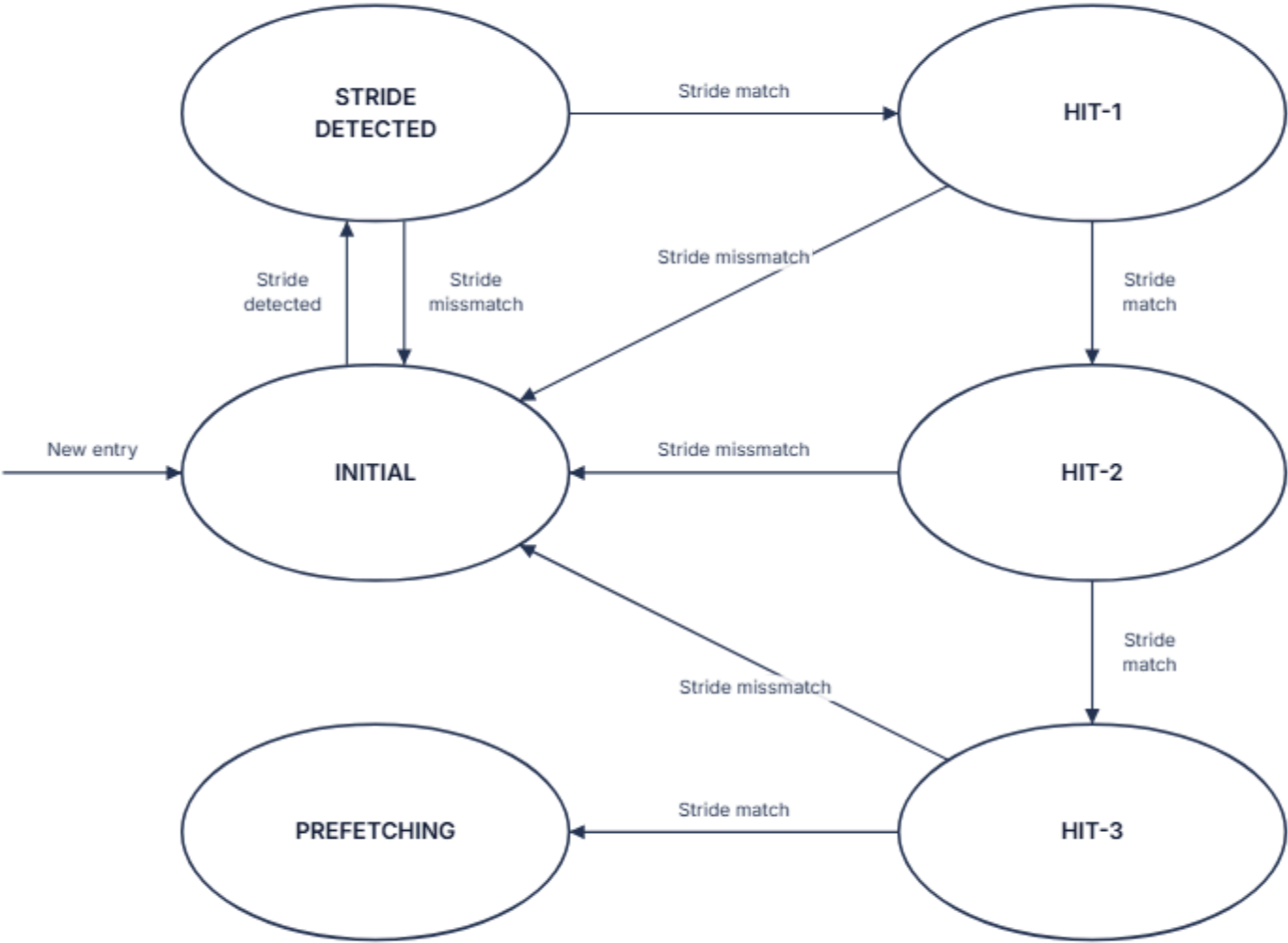
– Controller as FSM, state dependent outputs (Moore FSM) → breaking long paths



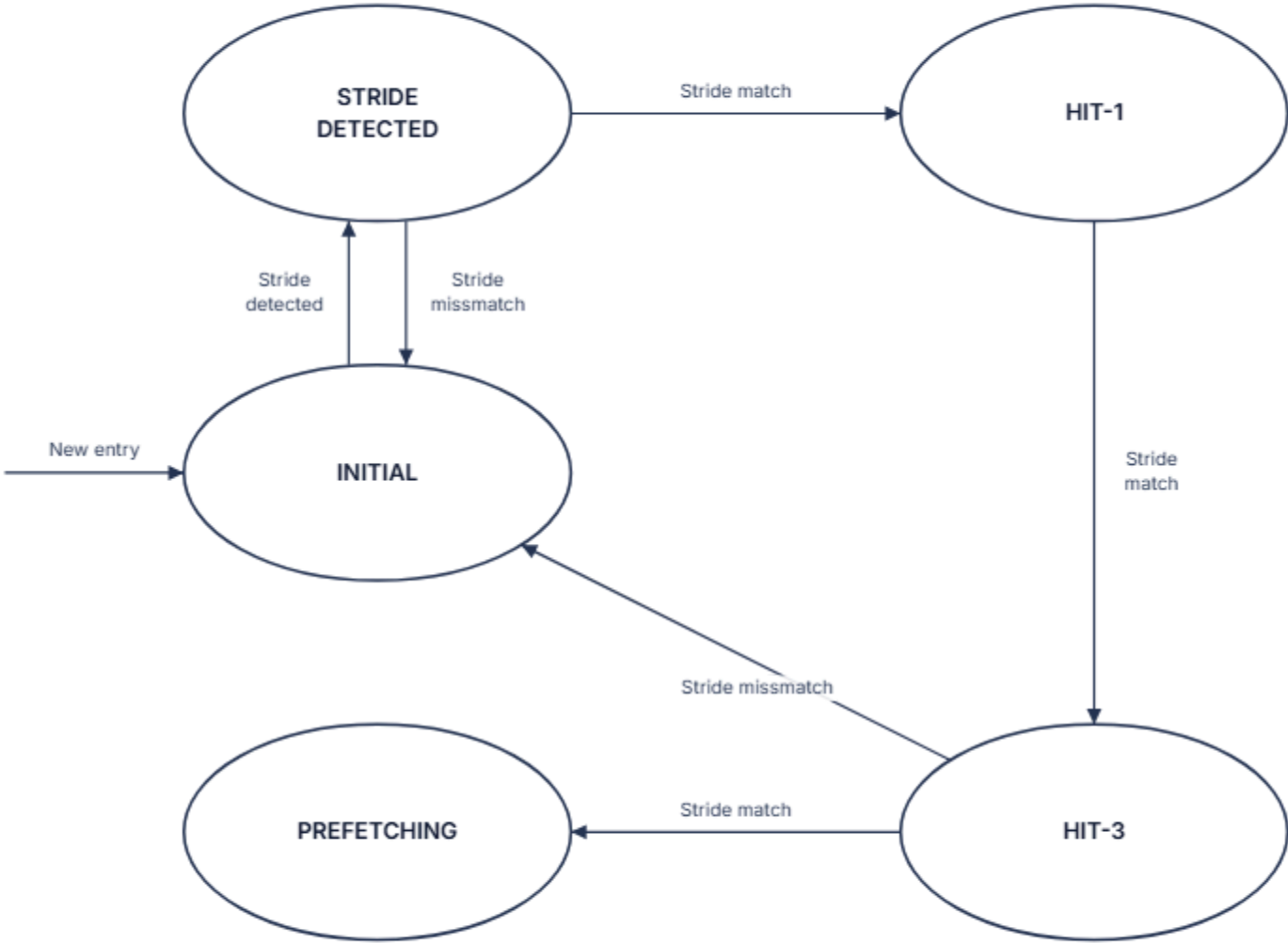
Data Prefetching: Fetchflare Integration

- Hiding memory latency by overlapping computation with memory access by speculatively accessing the data
- Integration of the FF — Modification of the FF FSM

Integration of the FF



Modification of the FF FSM



Experimental Setup & Methodology

- **Hardware Platform:** Digilent Genesys-2 development board with a Xilinx Kintex-7 FPGA.
- **Configurations Tested:** Master FPGA configuration and a minimized Small configuration.
- **Simulation:** Verilator was used for cycle-accurate open-source verification.
- **Benchmarks:** Evaluated using CoreMark, Dhrystone, and kernels adapted from PolyBench and STREAM.

CONFIGURATION PARAMETERS

PARAMETERS	FPGA CFG	SMALL CFG
Data Cache Size	8 KB	4 KB
Instruction Cache Size (B)	8 KB	8 KB
ICache and DCache Line Width	128	64
MMU present	1	0
A Extension	1	0
RVS and RVU	1	0
Store and Commit buffer depth	4	2
Debug Enabled	0	0



RESULTS & EVALUATION

Results: Timing & Frequency Analysis

- A

Master FPGA config
- B

Master Small config
- C

SB + ROB FPGA config
- D

SB + ROB Small config
- E

SB+ROB and COR FPGA config
- F

SB+ROB and COR Small config

CFG	SYNTH. FREQ (MHZ)	WNS (NS)	MAX FREQ (MHZ)	FAILING ENDPOINTS
A	130	-1.215	112.267	3880
B	140	-0.242	135.412	113
C	130	-0.5958	120.666	1205
D	140	-0.461	131.512	170
E	130	-0.092	128.763	29
F	140	-0.203	136.131	7



Results: Resource Utilization

- A

Master FPGA config
- B

Master Small config
- C

SB + ROB FPGA config
- D

SB + ROB Small config
- E

SB+ROB and COR FPGA config
- F

SB+ROB and COR Small config

CFG	LUT	FF	BRAM
A	9692	5410	13
B	7726	3809	7
C	9823	5569	13
D	7894	3961	7
E	10555	6099	13
F	8085	4119	7

TABLE 4.5 – RESOURCE COMPARISON

vs.	LUT	FF
A vs. C	1.35%	2.94%
B vs. D	2.1%	3.99%
A vs. E	8.90%	12.74%
B vs. F	4.65%	8.14%



RESULTS & EVALUATION

Results: Resource Utilization

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Master FPGA config
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RESULTS & EVALUATION

Results: Core Performance

- A

Master FPGA config
- B

Master Small config
- C

SB + ROB FPGA config
- D

SB + ROB Small config
- E

SB+ROB and COR FPGA config
- F

SB+ROB and COR Small config

CFG	MAX FREQ (MHZ)	COREMARK/MHZ	COREMARK	DHRYSTONE/MHZ
A	112.267	2.64	296.385	1747.03
B	135.412	2.64	357.488	1735.20
C	120.666	2.64	318.559	1747.03
D	131.512	2.64	347.192	1735.20
E	128.763	2.61	335.289	1740.03
F	136.131	2.61	355.302	1724.61



Results: Data Prefetching Performance

data prefetcher improved the cache miss by an average of 18.3% across various benchmarks.

18.3%

Average cache miss improvement from the data prefetcher across various benchmarks

Conclusion & Future Prospects

- Contribution: SB and Controller Opt brought 14% increase in the max frequency and a 13% increase in Coremark performance.
- Data prefetcher improved the cache miss by an average of 18.3% across various benchmarks.
- Future: the full advantage of the optimization needs to be evaluated on an ASIC design.
- Some optimizations, but not complete, can be listed as
 - Out-of-order issue of the instruction.
 - Extending with more parallel functional units on the execution stage with an available write-back port arbitrating with FIFO.
 - Load-store forwarding
 - Multi-level cache

+14%

Increase in max frequency from SB and Controller optimization

+13%

Increase in CoreMark performance

18.3%

Average cache miss improvement from the data prefetcher



— THANK YOU

Acknowledgement

Angela

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